

Qualification Pack



Junior Engineer -Electrical & Mechanical System (Metro Rail)

QP Code: DMR/LSC/Q0501

Version: 1.0

NSQF Level: 4.5

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DMR/LSC/Q0501: Junior Engineer -Electrical & Mechanical System (Metro Rail)

Brief Job Description

A Junior Engineer -Electrical & Mechanical Systems (Metro Rail) is responsible for the operation and maintenance of Electro-Mechanical (E&M) systems in metro stations, including underground and elevated setups. The individual ensures energy efficiency, infrastructure management, and compliance with safety standards for escalators and elevators, contributing to smooth station operations. The person coordinates with teams to implement preventive maintenance and address technical issues.

Personal Attributes

The individual should possess strong analytical and organizational skills, and attention to detail. A proactive approach to problem-solving and commitment to safety and efficiency are essential.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DMR/LSC/N9901: Implement metro operations training and safety procedures](#)
2. [DMR/LSC/N0501: Monitor the operations and maintenance of E&M systems in metro railways](#)
3. [DMR/LSC/N0502: Monitor and maintain the underground and elevated station systems](#)
4. [DMR/LSC/N0503: Monitor energy utilization and infrastructure maintenance in metro stations](#)
5. [DMR/LSC/N0504: Monitor and ensure safe escalator and elevator operations in metro systems](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Logistics
Sub-Sector	Transport (Metro Rail)
Occupation	Electrical & Mechanical
Country	India
NSQF Level	4.5

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Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2153.9900
Minimum Educational Qualification & Experience	Diploma (3 years Engineering Diploma* / equivalent Electrical / Mechanical or in relevant field)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-4.5-TW-04144-2025-V1-DMRCA
NQR Version	1.0

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DMR/LSC/N9901: Implement metro operations training and safety procedures

Description

This unit focuses on implementing structured training and safety measures for metro operations, ensuring technical knowledge, system readiness, and emergency preparedness.

Scope

The scope covers the following :

- Training Module Overview
- Safety Overview
- Store Management and Procurement
- Overview of Telecom, Signaling, and Track Systems
- Overview of Traction, SCADA, and E&M Systems
- OCC and Rolling Stock Overview
- Operations and Customer Care
- Practical Visits
- First Aid Training
- Fire Fighting Systems
- Security in Metro Systems
- Metro Rail General Rules 2020
- Safety Circulars and Contingency Plans

Elements and Performance Criteria

Training Module Overview

To be competent, the user/individual on the job must be able to:

- PC1.** follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.
- PC2.** conduct identity verification and maintain attendance records
- PC3.** ensure adherence to training timelines and program deliverables

Safety Overview

To be competent, the user/individual on the job must be able to:

- PC4.** identify the role of the safety department and follow measures for personal, passenger, and system safety
- PC5.** use appropriate Personal Protective Equipment (PPE) in all designated areas
- PC6.** follow the hazard identification and risk assessment processes

Store Management and Procurement

To be competent, the user/individual on the job must be able to:

- PC7.** perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.
- PC8.** monitor inventory levels and ensure timely requisitions for required materials
- PC9.** maintain records of procurement processes and vendor communication

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Overview of Telecom, Signaling, and Track Systems

To be competent, the user/individual on the job must be able to:

- PC10.** identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks
- PC11.** identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.
- PC12.** identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks
- PC13.** report malfunctions in telecom, signaling, or track systems using the correct reporting protocols

Overview of Traction, SCADA, and E&M Systems

To be competent, the user/individual on the job must be able to:

- PC14.** identify traction power and SCADA components and interpret sectioning diagrams
- PC15.** identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality
- PC16.** report unusual system performance and assist in routine inspections

OCC and Rolling Stock Overview

To be competent, the user/individual on the job must be able to:

- PC17.** familiarize with OCC systems, tools, and the role of OCC controllers
- PC18.** identify rolling stock components and their maintenance procedures
- PC19.** report rolling stock irregularities observed during operations

Operations and Customer Care

To be competent, the user/individual on the job must be able to:

- PC20.** familiarize with station operations, revenue management, and crew control functions
- PC21.** identify public complaint channels, possible causes, and resolve complaints using proper processes
- PC22.** assist in managing passenger inquiries and ensuring customer satisfaction

Practical Visits

To be competent, the user/individual on the job must be able to:

- PC23.** conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations
- PC24.** document key observations during practical visits for knowledge reinforcement
- PC25.** identify operational challenges during visits and suggest improvement areas

First Aid Training

To be competent, the user/individual on the job must be able to:

- PC26.** identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)
- PC27.** follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness
- PC28.** perform artificial respiration, CPR, and safe transportation of injured persons using stretchers
- PC29.** participate in simulated first aid scenarios to enhance preparedness

Fire Fighting Systems

To be competent, the user/individual on the job must be able to:

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- PC30.** identify elements of combustion, classes of fire, and appropriate extinguishing methods
- PC31.** operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants
- PC32.** conduct practical drills and identify fire suppression equipment at metro stations
- PC33.** respond to fire incidents promptly, ensuring passenger and system safety

Security in Metro Systems

To be competent, the user/individual on the job must be able to:

- PC34.** identify security threats, red alert situations, and follow protocols for disaster management
- PC35.** handle theft cases, manage abandoned persons, and lodge FIRs as per protocol
- PC36.** coordinate with security personnel to mitigate security breaches effectively

Metro Rail General Rules 2020

To be competent, the user/individual on the job must be able to:

- PC37.** interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols
- PC38.** follow operational procedures for maintenance, station control duties, and driverless train systems
- PC39.** ensure compliance with rules for safe and efficient train operations

Safety Circulars and Contingency Plans

To be competent, the user/individual on the job must be able to:

- PC40.** implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation
- PC41.** follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises
- PC42.** ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods
- PC43.** manage degraded mode operations, including train movement during interlocking failures
- PC44.** oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections
- PC45.** facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks
- PC46.** execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras
- PC47.** oversee train stabling, depot entry/exit, train movement, and fitness checks
- PC48.** conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detainment of passengers due to technical failure, explosion, or fire
- PC49.** operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train
- PC50.** coordinate vehicle movement on link lines and rescue immobilized trains
- PC51.** respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety
- PC52.** operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of punctuality, discipline, and structured training
- KU2.** the procedures for trainee verification and attendance record maintenance
- KU3.** the role of time management in achieving training milestones
- KU4.** the role of the safety department and station/depot safety protocols
- KU5.** the use and importance of PPE
- KU6.** the basics of hazard identification, risk management, and accident reporting
- KU7.** the basics of stores management (inventory control, documentation)
- KU8.** the procurement processes and coordination across departments
- KU9.** the procedures for vendor management and maintaining procurement transparency
- KU10.** the telecom components, including AFC, CCTV, clocks, and communication systems
- KU11.** the basics of signaling systems (CATC, CBTC) and reporting unusual signal observations
- KU12.** the key P-Way components, track patrolling, and reporting track conditions
- KU13.** the common malfunctions in telecom, signaling, and track systems
- KU14.** the traction power systems and sectioning diagrams
- KU15.** the importance of SCADA and its functionalities
- KU16.** the role of E&M systems (DG sets, UPS) and their use during emergencies
- KU17.** the key preventive maintenance checks for E&M systems
- KU18.** the OCC systems, tools, and controller responsibilities
- KU19.** the rolling stock components, features, and basic maintenance
- KU20.** the reporting and addressing of common rolling stock issues
- KU21.** the station operations, revenue management, and crew coordination
- KU22.** the causes of public complaints and grievance resolution processes
- KU23.** the techniques for effective customer service and passenger engagement
- KU24.** the observations and learning outcomes from OCC, station, depot, and tunnel visits
- KU25.** the identifying real-world challenges during visits and linking theory to practice
- KU26.** the basic principles of first aid and key human body functions
- KU27.** the management of bleeding, fractures, burns, and unconsciousness
- KU28.** the artificial respiration, CPR, and safe patient transportation
- KU29.** the importance of first aid preparedness during emergencies
- KU30.** the elements of combustion, fire classifications, and extinguishing methods
- KU31.** the principles of passive/active fire protection systems
- KU32.** the practical usage of fire extinguishers, hydrants, and drills
- KU33.** the emergency response procedures during fire incidents
- KU34.** the security protocols, red alert situations, and theft management processes
- KU35.** the protocols for managing abandoned persons and disaster situations
- KU36.** the role of security personnel in ensuring passenger and asset safety
- KU37.** the key terminologies, signal usage, and operational protocols

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- KU38.** the rules for maintenance works, train operations, and driverless train systems
- KU39.** the role of regulatory compliance in metro rail safety
- KU40.** the safety circulars and contingency plans for fire/security incidents
- KU41.** the importance of proactive contingency planning for emergency scenarios.
- KU42.** the importance role-playing and drills to improve safety and response performance
- KU43.** the procedures for TSR imposition, axle counter reset, Train ID management, and CATS operations
- KU44.** the engineering possession protocols for maintenance tasks during non-revenue hours, including solar system testing and hot work
- KU45.** the guidelines for safely halting train operations during emergencies, natural disasters, or technical failures
- KU46.** the steps for managing degraded mode operations and interlocking failures
- KU47.** the SOP and SWO/DWO lifecycle management, including revisions and timetable trials for new sections
- KU48.** the ATO, UTO, and ATB mode operations, emphasizing operational readiness and AFTC checks
- KU49.** the procedures for neutral section negotiation, pantograph damage management, and monitoring Panto-OHE interfaces using cameras
- KU50.** the best practices for train stabling, depot operations, and train fitness verification
- KU51.** the emergency evacuation protocols, focusing on passenger safety in UG and mid-tunnel situations
- KU52.** the guidelines for operating non-ATP vehicles, handling rail fractures, weld failures, and resetting alarms during rollbacks
- KU53.** the vehicle operations on link lines and immobilized train rescue procedures
- KU54.** the protocols for fire/smoke incidents, particularly UG fires, and ensuring safety during such incidents
- KU55.** the operation of ETS/BLS plungers and removal of foreign objects from OHE systems, along with OHE safety practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey clear and concise information to teams, junior engineers, and customers
- GS2.** effectively prioritize tasks and adhere to training schedules and operational timelines
- GS3.** identify operational challenges and propose practical solutions during training or real-time operations
- GS4.** assess system performance, identify malfunctions, and report anomalies accurately
- GS5.** work cohesively with peers, instructors, and junior engineers to meet training objectives
- GS6.** address passenger queries and manage grievances professionally
- GS7.** proactively identify risks, follow safety protocols, and contribute to hazard-free operations
- GS8.** adapt to diverse training environments, system layouts, and evolving operational scenarios
- GS9.** accurately maintain logs, records, and reports for training and practical activities

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- GS10.** monitor systems, equipment, and processes, identifying areas requiring intervention or improvement

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Training Module Overview</i>	2	2	-	1
PC1. follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.	-	-	-	-
PC2. conduct identity verification and maintain attendance records	-	-	-	-
PC3. ensure adherence to training timelines and program deliverables	-	-	-	-
<i>Safety Overview</i>	2	2	-	1
PC4. identify the role of the safety department and follow measures for personal, passenger, and system safety	-	-	-	-
PC5. use appropriate Personal Protective Equipment (PPE) in all designated areas	-	-	-	-
PC6. follow the hazard identification and risk assessment processes	-	-	-	-
<i>Store Management and Procurement</i>	2	2	-	1
PC7. perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.	-	-	-	-
PC8. monitor inventory levels and ensure timely requisitions for required materials	-	-	-	-
PC9. maintain records of procurement processes and vendor communication	-	-	-	-
<i>Overview of Telecom, Signaling, and Track Systems</i>	2	2	-	1
PC10. identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks	-	-	-	-
PC11. identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks	-	-	-	-
PC13. report malfunctions in telecom, signaling, or track systems using the correct reporting protocols	-	-	-	-
<i>Overview of Traction, SCADA, and E&M Systems</i>	2	2	-	1
PC14. identify traction power and SCADA components and interpret sectioning diagrams	-	-	-	-
PC15. identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality	-	-	-	-
PC16. report unusual system performance and assist in routine inspections	-	-	-	-
<i>OCC and Rolling Stock Overview</i>	2	2	-	1
PC17. familiarize with OCC systems, tools, and the role of OCC controllers	-	-	-	-
PC18. identify rolling stock components and their maintenance procedures	-	-	-	-
PC19. report rolling stock irregularities observed during operations	-	-	-	-
<i>Operations and Customer Care</i>	2	2	-	1
PC20. familiarize with station operations, revenue management, and crew control functions	-	-	-	-
PC21. identify public complaint channels, possible causes, and resolve complaints using proper processes	-	-	-	-
PC22. assist in managing passenger inquiries and ensuring customer satisfaction	-	-	-	-
<i>Practical Visits</i>	1	1	-	0.5
PC23. conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. document key observations during practical visits for knowledge reinforcement	-	-	-	-
PC25. identify operational challenges during visits and suggest improvement areas	-	-	-	-
<i>First Aid Training</i>	1	1	-	0.5
PC26. identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)	-	-	-	-
PC27. follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness	-	-	-	-
PC28. perform artificial respiration, CPR, and safe transportation of injured persons using stretchers	-	-	-	-
PC29. participate in simulated first aid scenarios to enhance preparedness	-	-	-	-
<i>Fire Fighting Systems</i>	1	1	-	0.5
PC30. identify elements of combustion, classes of fire, and appropriate extinguishing methods	-	-	-	-
PC31. operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants	-	-	-	-
PC32. conduct practical drills and identify fire suppression equipment at metro stations	-	-	-	-
PC33. respond to fire incidents promptly, ensuring passenger and system safety	-	-	-	-
<i>Security in Metro Systems</i>	1	1	-	0.5
PC34. identify security threats, red alert situations, and follow protocols for disaster management	-	-	-	-
PC35. handle theft cases, manage abandoned persons, and lodge FIRs as per protocol	-	-	-	-
PC36. coordinate with security personnel to mitigate security breaches effectively	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Metro Rail General Rules 2020</i>	1	1	-	0.5
PC37. interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols	-	-	-	-
PC38. follow operational procedures for maintenance, station control duties, and driverless train systems	-	-	-	-
PC39. ensure compliance with rules for safe and efficient train operations	-	-	-	-
<i>Safety Circulars and Contingency Plans</i>	1	1	-	0.5
PC40. implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation	-	-	-	-
PC41. follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises	-	-	-	-
PC42. ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods	-	-	-	-
PC43. manage degraded mode operations, including train movement during interlocking failures	-	-	-	-
PC44. oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections	-	-	-	-
PC45. facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks	-	-	-	-
PC46. execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC47. oversee train stabling, depot entry/exit, train movement, and fitness checks	-	-	-	-
PC48. conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire	-	-	-	-
PC49. operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train	-	-	-	-
PC50. coordinate vehicle movement on link lines and rescue immobilized trains	-	-	-	-
PC51. respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety	-	-	-	-
PC52. operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines	-	-	-	-
NOS Total	20	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N9901
NOS Name	Implement metro operations training and safety procedures
Sector	Logistics
Sub-Sector	
Occupation	Foundation – Metro (Rail)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DMR/LSC/N0501: Monitor the operations and maintenance of E&M systems in metro railways

Description

This unit is about the monitoring, troubleshooting, and optimizing critical Electro-Mechanical (E&M) systems, including ECS, TVS, fire suppression, and hydraulic systems, for metro operations, ensuring efficient operations, compliance with codal requirements, and troubleshooting during commissioning.

Scope

The scope covers the following :

- Overview of E&M Systems
- Introduction to Variable Refrigerant Volume (VRV) System
- Site Visit to Elevated Station for VRV System
- Environmental Control System (ECS)
- Tunnel Ventilation System (TVS)
- Visit to ECS and TVS Demo Room
- Site Visit to Underground Station for ECS System
- Fire Detection and Suppression Systems
- Visit to Underground Station for TVS and Fire Equipment
- Hydraulic Systems
- Site Visit to PTM
- Station Illumination
- Building Management System (BMS)

Elements and Performance Criteria

Overview of E&M Systems

To be competent, the user/individual on the job must be able to:

- PC1.** check the earthing and lightning protection systems at metro stations, ensuring compliance with DMRC guidelines
- PC2.** monitor the installation and maintenance of earthing and lightning protection systems to avoid operational disruptions

Introduction to Variable Refrigerant Volume (VRV) System

To be competent, the user/individual on the job must be able to:

- PC3.** identify the functions of VRV system components, including indoor and outdoor units
- PC4.** troubleshoot VRV system cards and recommend solutions to improve performance

Site Visit to Elevated Station for VRV System

To be competent, the user/individual on the job must be able to:

- PC5.** evaluate the VRV system operation on-site at elevated metro stations to identify operational issues
- PC6.** document findings and propose corrective actions for optimal VRV system performance

Environmental Control System (ECS)

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To be competent, the user/individual on the job must be able to:

PC7. identify the refrigeration cycle and functions of ECS components, including chillers and cooling towers

PC8. identify commissioning issues in ECS installations and recommend effective solutions

Tunnel Ventilation System (TVS)

To be competent, the user/individual on the job must be able to:

PC9. check the operational modes and codal compliances for tunnel ventilation systems

PC10. address interface challenges during TVS commissioning to ensure seamless operation

Visit to ECS and TVS Demo Room

To be competent, the user/individual on the job must be able to:

PC11. observe ECS and TVS demo operations and identify common challenges during system commissioning

PC12. record observations and apply learning to enhance practical understanding of ECS and TVS systems

Site Visit to Underground Station for ECS System

To be competent, the user/individual on the job must be able to:

PC13. conduct detailed inspections of ECS systems at underground stations to evaluate their functionality

PC14. collaborate with the team to resolve identified ECS issues and improve operational efficiency

Fire Detection and Suppression Systems

To be competent, the user/individual on the job must be able to:

PC15. examine fire detection and suppression systems for adherence to codal requirements in metro stations

PC16. identify commissioning challenges in fire systems, particularly in tunnels, and recommend improvements

Visit to Underground Station for TVS and Fire Equipment

To be competent, the user/individual on the job must be able to:

PC17. check the TVS and fire systems on-site to ensure compliance and functionality in underground stations

PC18. document and address discrepancies in TVS and fire systems for improved safety measures

Hydraulic Systems

To be competent, the user/individual on the job must be able to:

PC19. check the water supply and wastewater disposal systems for proper operation and maintenance

PC20. analyze case studies of hydraulic system failures during commissioning and suggest preventive measures

Site Visit to PTM

To be competent, the user/individual on the job must be able to:

PC21. check Pump Testing Machines (PTM) during site visits to verify operational efficiency

PC22. identify maintenance needs and recommend best practices for PTM management

Station Illumination

To be competent, the user/individual on the job must be able to:

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- PC23.** assess the illumination levels at metro stations and ensure they meet percentage lighting requirements
- PC24.** implement energy-efficient solutions to enhance station illumination while reducing operational costs

Building Management System (BMS)

To be competent, the user/individual on the job must be able to:

- PC25.** check the operation of Building Management Systems, including NT-SCADA features like alarms, events, and trends
- PC26.** monitor system backups and ensure accurate report generation for BMS performance analysis

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the key principles of earthing and lightning protection in metro railway systems
- KU2.** the components and functioning of VRV systems and their applications in elevated metro stations
- KU3.** the basics of the refrigeration cycle, ECS components, and differences between water-cooled and air-cooled chillers
- KU4.** the design and operational principles of tunnel ventilation systems, including codal compliance and interface management
- KU5.** the functional knowledge of fire detection and suppression systems, including commissioning challenges in tunnels
- KU6.** the basics of hydraulic systems, water supply, and wastewater disposal with examples of real-world failures
- KU7.** the concepts of station illumination, energy efficiency, and percentage lighting distribution
- KU8.** the features and applications of Building Management Systems, including NT-SCADA functionalities and reporting tools
- KU9.** the common commissioning challenges across E&M systems and practical troubleshooting methods
- KU10.** the importance of codal compliances and regulatory standards for metro E&M systems

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** clearly convey technical information to team members and stakeholders
- GS2.** assess system performance critically to identify areas of improvement
- GS3.** address commissioning and operational challenges with effective solutions
- GS4.** ensure adherence to codal requirements for E&M systems
- GS5.** coordinate effectively with multidisciplinary teams during inspections and commissioning
- GS6.** maintain precise records of system inspections and findings
- GS7.** follow advancements in metro rail E&M technologies and systems

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- GS8.** plan and prioritize tasks to meet project deadlines efficiently
- GS9.** evaluate operational strategies to improve efficiency and reliability
- GS10.** focus on implementing safe practices while managing E&M systems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Overview of E&M Systems</i>	1	1	-	0.5
PC1. check the earthing and lightning protection systems at metro stations, ensuring compliance with DMRC guidelines	-	-	-	-
PC2. monitor the installation and maintenance of earthing and lightning protection systems to avoid operational disruptions	-	-	-	-
<i>Introduction to Variable Refrigerant Volume (VRV) System</i>	2	2	-	1
PC3. identify the functions of VRV system components, including indoor and outdoor units	-	-	-	-
PC4. troubleshoot VRV system cards and recommend solutions to improve performance	-	-	-	-
<i>Site Visit to Elevated Station for VRV System</i>	1	1	-	1
PC5. evaluate the VRV system operation on-site at elevated metro stations to identify operational issues	-	-	-	-
PC6. document findings and propose corrective actions for optimal VRV system performance	-	-	-	-
<i>Environmental Control System (ECS)</i>	2	2	-	1
PC7. identify the refrigeration cycle and functions of ECS components, including chillers and cooling towers	-	-	-	-
PC8. identify commissioning issues in ECS installations and recommend effective solutions	-	-	-	-
<i>Tunnel Ventilation System (TVS)</i>	2	2	-	1
PC9. check the operational modes and codal compliances for tunnel ventilation systems	-	-	-	-
PC10. address interface challenges during TVS commissioning to ensure seamless operation	-	-	-	-
<i>Visit to ECS and TVS Demo Room</i>	1	1	-	0.5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. observe ECS and TVS demo operations and identify common challenges during system commissioning	-	-	-	-
PC12. record observations and apply learning to enhance practical understanding of ECS and TVS systems	-	-	-	-
<i>Site Visit to Underground Station for ECS System</i>	1	1	-	0.5
PC13. conduct detailed inspections of ECS systems at underground stations to evaluate their functionality	-	-	-	-
PC14. collaborate with the team to resolve identified ECS issues and improve operational efficiency	-	-	-	-
<i>Fire Detection and Suppression Systems</i>	2	2	-	1
PC15. examine fire detection and suppression systems for adherence to codal requirements in metro stations	-	-	-	-
PC16. identify commissioning challenges in fire systems, particularly in tunnels, and recommend improvements	-	-	-	-
<i>Visit to Underground Station for TVS and Fire Equipment</i>	1	1	-	0.5
PC17. check the TVS and fire systems on-site to ensure compliance and functionality in underground stations	-	-	-	-
PC18. document and address discrepancies in TVS and fire systems for improved safety measures	-	-	-	-
<i>Hydraulic Systems</i>	2	2	-	1
PC19. check the water supply and wastewater disposal systems for proper operation and maintenance	-	-	-	-
PC20. analyze case studies of hydraulic system failures during commissioning and suggest preventive measures	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Site Visit to PTM</i>	1	1	-	0.5
PC21. check Pump Testing Machines (PTM) during site visits to verify operational efficiency	-	-	-	-
PC22. identify maintenance needs and recommend best practices for PTM management	-	-	-	-
<i>Station Illumination</i>	2	2	-	0.5
PC23. assess the illumination levels at metro stations and ensure they meet percentage lighting requirements	-	-	-	-
PC24. implement energy-efficient solutions to enhance station illumination while reducing operational costs	-	-	-	-
<i>Building Management System (BMS)</i>	2	2	-	1
PC25. check the operation of Building Management Systems, including NT-SCADA features like alarms, events, and trends	-	-	-	-
PC26. monitor system backups and ensure accurate report generation for BMS performance analysis	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0501
NOS Name	Monitor the operations and maintenance of E&M systems in metro railways
Sector	Logistics
Sub-Sector	
Occupation	Electrical & Mechanical
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0502: Monitor and maintain the underground and elevated station systems

Description

This unit outlines the competencies required to monitor and maintain E&M systems like BMS, UPS, DG, LT power distribution, and emergency tools in both elevated and underground metro stations, ensuring safe, efficient, and reliable operations.

Scope

The scope covers the following :

- Site Visit to Underground Stations for BMS and NTS
- Familiarization with UPS Systems
- Familiarization with DG Systems in Underground Stations
- Site Visit for DG and UPS Systems in Underground Stations
- LT Power Distribution at Underground/RC Stations
- Site Visit to Elevated Stations
- Night Visit for Maintenance Procedures
- Operation of ASC at OCC/SHPK
- Engineers Roles and Responsibilities
- Review of Maintenance Schedules and Procedures
- Review of Mock Drills
- Site Visit for Tools Inspection
- Failure Analysis of E&M Systems
- Site Visit to Tunnel Training Centre

Elements and Performance Criteria

Site Visit to Underground Stations for BMS and NTS

To be competent, the user/individual on the job must be able to:

- PC1.** conduct visits to underground stations to assess Building Management System (BMS) functionality and ensure adherence to operational standards
- PC2.** verify the integration and effectiveness of Network Time Synchronization (NTS) in supporting BMS operations

Familiarization with UPS Systems

To be competent, the user/individual on the job must be able to:

- PC3.** identify the types and working principles of Uninterruptible Power Supply (UPS) systems used in metro stations
- PC4.** follow the procedure for paralleling UPS systems and batteries for continuous power supply

Familiarization with DG Systems in Underground Stations

To be competent, the user/individual on the job must be able to:

- PC5.** familiarize with the parts, functionality, and interlocking of Diesel Generator (DG) systems with Auxiliary Substations (ASS)
- PC6.** identify general problems in DG systems and follow corrective maintenance procedures

Qualification Pack

Site Visit for DG and UPS Systems in Underground Stations

To be competent, the user/individual on the job must be able to:

- PC7.** conduct a detailed site visit to inspect the DG systems at underground stations for operational efficiency
- PC8.** check the UPS system's performance and maintenance records for compliance with safety standards

LT Power Distribution at Underground/RC Stations

To be competent, the user/individual on the job must be able to:

- PC9.** identify different types of loads in LT power distribution systems and their requirements
- PC10.** familiarize with the ACB interlocking schemes in different operational phases and review relay settings for discrimination

Site Visit to Elevated Stations

To be competent, the user/individual on the job must be able to:

- PC11.** check the elevated station power distribution systems, DG systems, and fire protection mechanisms
- PC12.** evaluate the functionality of pumps, UPS systems, and lighting systems during operational hours

Night Visit for Maintenance Procedures

To be competent, the user/individual on the job must be able to:

- PC13.** conduct night visits to observe and ensure adherence to maintenance procedures in elevated systems
- PC14.** verify the documentation and adherence to maintenance schedules during night inspections

Operation of ASC at OCC/SHPK

To be competent, the user/individual on the job must be able to:

- PC15.** familiarize with the role of Automatic Supervisory Control (ASC) during tunnel ventilation system operation
- PC16.** implement appropriate measures during SCADA communication failures to ensure operational continuity

Engineers' Roles and Responsibilities

To be competent, the user/individual on the job must be able to:

- PC17.** familiarize with the roles and responsibilities of junior engineers and staff for E&M, ECS, and fire protection systems
- PC18.** maintain and review records to ensure proper upkeep of maintenance logs and inspection schedules

Review of Maintenance Schedules and Procedures

To be competent, the user/individual on the job must be able to:

- PC19.** evaluate the adherence to maintenance schedules, Special Maintenance Instructions (SMI), and Standard Operating Instructions (SOI)
- PC20.** maintain regular and detailed records and conduct regular checks of E&M works

Review of Mock Drills

To be competent, the user/individual on the job must be able to:

- PC21.** review the effectiveness of various E&M mock drills and document lessons learned
- PC22.** maintain a record of tools used during mock drills to ensure readiness for emergencies

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Site Visit for Tools Inspection

To be competent, the user/individual on the job must be able to:

- PC23.** conduct site visits to underground stations to inspect the condition and availability of tools required for E&M operations
- PC24.** verify the compliance of tools with operational standards and safety guidelines

Failure Analysis of E&M Systems

To be competent, the user/individual on the job must be able to:

- PC25.** analyze failures in E&M systems and identify root causes to prevent recurrence
- PC26.** implement corrective actions and prepare detailed reports on failure analysis outcomes

Site Visit to Tunnel Training Centre

To be competent, the user/individual on the job must be able to:

- PC27.** conduct visits to the tunnel training center to observe the operation of E&M systems in tunnels
- PC28.** assess the functionality of trackway sprinkler systems and other emergency services

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the Building Management System (BMS) structure, components, and functionalities
- KU2.** the Network Time Systems (NTS) synchronization principles and applications in metro systems
- KU3.** the types, working principles, paralleling methods, and battery maintenance of UPS systems
- KU4.** the components, interlocking mechanisms, common failures, and maintenance practices for DG Systems
- KU5.** LT power schemes, load types, ACB interlocking, relay settings, and discrimination techniques
- KU6.** the operational procedures for elevated and underground station systems
- KU7.** the maintenance procedures, documentation, and compliance standards
- KU8.** the functions of ASC systems and SCADA communication protocols
- KU9.** the emergency response protocols, including mock drill execution and analysis
- KU10.** the tools and equipment standards for E&M operations
- KU11.** the failure analysis techniques and corrective maintenance processes

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze system failures and identify practical solutions
- GS2.** effectively convey instructions and feedback to staff
- GS3.** address operational challenges with efficient solutions
- GS4.** collaborate with cross-functional teams for system maintenance
- GS5.** maintain accurate maintenance logs and inspection records

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Qualification Pack

- GS6.** make informed decisions during emergencies or failures
- GS7.** ensure all systems comply with operational standards
- GS8.** prioritize and execute tasks within designated schedules
- GS9.** oversee teams effectively during routine and emergency operations
- GS10.** respond flexibly to dynamic operational demands and emergencies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Site Visit to Underground Stations for BMS and NTS</i>	1	1	-	1
PC1. conduct visits to underground stations to assess Building Management System (BMS) functionality and ensure adherence to operational standards	-	-	-	-
PC2. verify the integration and effectiveness of Network Time Synchronization (NTS) in supporting BMS operations	-	-	-	-
<i>Familiarization with UPS Systems</i>	2	2	-	1
PC3. identify the types and working principles of Uninterruptible Power Supply (UPS) systems used in metro stations	-	-	-	-
PC4. follow the procedure for paralleling UPS systems and batteries for continuous power supply	-	-	-	-
<i>Familiarization with DG Systems in Underground Stations</i>	1	1	-	1
PC5. familiarize with the parts, functionality, and interlocking of Diesel Generator (DG) systems with Auxiliary Substations (ASS)	-	-	-	-
PC6. identify general problems in DG systems and follow corrective maintenance procedures	-	-	-	-
<i>Site Visit for DG and UPS Systems in Underground Stations</i>	1	1	-	1
PC7. conduct a detailed site visit to inspect the DG systems at underground stations for operational efficiency	-	-	-	-
PC8. check the UPS system's performance and maintenance records for compliance with safety standards	-	-	-	-
<i>LT Power Distribution at Underground/RC Stations</i>	1	1	-	1
PC9. identify different types of loads in LT power distribution systems and their requirements	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. familiarize with the ACB interlocking schemes in different operational phases and review relay settings for discrimination	-	-	-	-
<i>Site Visit to Elevated Stations</i>	2	2	-	1
PC11. check the elevated station power distribution systems, DG systems, and fire protection mechanisms	-	-	-	-
PC12. evaluate the functionality of pumps, UPS systems, and lighting systems during operational hours	-	-	-	-
<i>Night Visit for Maintenance Procedures</i>	2	2	-	0.5
PC13. conduct night visits to observe and ensure adherence to maintenance procedures in elevated systems	-	-	-	-
PC14. verify the documentation and adherence to maintenance schedules during night inspections	-	-	-	-
<i>Operation of ASC at OCC/SHPK</i>	2	2	-	0.5
PC15. familiarize with the role of Automatic Supervisory Control (ASC) during tunnel ventilation system operation	-	-	-	-
PC16. implement appropriate measures during SCADA communication failures to ensure operational continuity	-	-	-	-
<i>Engineers' Roles and Responsibilities</i>	2	2	-	0.5
PC17. familiarize with the roles and responsibilities of junior engineers and staff for E&M, ECS, and fire protection systems	-	-	-	-
PC18. maintain and review records to ensure proper upkeep of maintenance logs and inspection schedules	-	-	-	-
<i>Review of Maintenance Schedules and Procedures</i>	1	1	-	0.5
PC19. evaluate the adherence to maintenance schedules, Special Maintenance Instructions (SMI), and Standard Operating Instructions (SOI)	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. maintain regular and detailed records and conduct regular checks of E&M works	-	-	-	-
<i>Review of Mock Drills</i>	1	1	-	0.5
PC21. review the effectiveness of various E&M mock drills and document lessons learned	-	-	-	-
PC22. maintain a record of tools used during mock drills to ensure readiness for emergencies	-	-	-	-
<i>Site Visit for Tools Inspection</i>	1	1	-	0.5
PC23. conduct site visits to underground stations to inspect the condition and availability of tools required for E&M operations	-	-	-	-
PC24. verify the compliance of tools with operational standards and safety guidelines	-	-	-	-
<i>Failure Analysis of E&M Systems</i>	2	2	-	0.5
PC25. analyze failures in E&M systems and identify root causes to prevent recurrence	-	-	-	-
PC26. implement corrective actions and prepare detailed reports on failure analysis outcomes	-	-	-	-
<i>Site Visit to Tunnel Training Centre</i>	1	1	-	0.5
PC27. conduct visits to the tunnel training center to observe the operation of E&M systems in tunnels	-	-	-	-
PC28. assess the functionality of trackway sprinkler systems and other emergency services	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0502
NOS Name	Monitor and maintain the underground and elevated station systems
Sector	Logistics
Sub-Sector	
Occupation	Electrical & Mechanical
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0503: Monitor energy utilization and infrastructure maintenance in metro stations

Description

This unit outlines the procedures for monitoring and maintaining E&M systems in elevated and underground metro stations, to ensure the efficient operation and maintenance of power, cooling, and environmental systems, emphasizing energy conservation and safety compliance.

Scope

The scope covers the following :

- Night Visits for Maintenance Activities in Underground Stations
- Energy Conservation and Management Practices
- Power Supply Distribution and Cooling Systems
- Earthing, Water Treatment, and Pumping Systems
- Environmental Control Systems (ECS) and Tunnel Ventilation Systems (TVS)
- Thermal Energy Storage Systems

Elements and Performance Criteria

Night Visits for Maintenance Activities in Underground Stations

To be competent, the user/individual on the job must be able to:

- PC1.** check ongoing E&M maintenance activities during night visits to ensure adherence to standard operating procedures and safety regulations
- PC2.** verify the functionality and maintenance status of critical systems such as power distribution, lighting, ventilation, and escalators

Energy Conservation and Management Practices

To be competent, the user/individual on the job must be able to:

- PC3.** monitor and analyze energy consumption trends to identify potential areas for conservation
- PC4.** follow appropriate energy-saving measures without compromising operational efficiency

Power Supply Distribution and Cooling Systems

To be competent, the user/individual on the job must be able to:

- PC5.** evaluate the power supply distribution scheme at stations, ensuring compliance with DMRC standards and safety protocols
- PC6.** check the performance and maintenance of air-cooled chillers and aerosol fire suppression systems

Earthing, Water Treatment, and Pumping Systems

To be competent, the user/individual on the job must be able to:

- PC7.** check earthing systems and ensure proper grounding to mitigate electrical hazards
- PC8.** coordinate the maintenance of water treatment plants (WTP), water supply pumps, seepage, and sewage systems

Environmental Control Systems (ECS) and Tunnel Ventilation Systems (TVS)

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC9.** check ECS and TVS in underground stations for effective temperature and air quality management
- PC10.** ensure the ECS and TVS systems meet operational requirements and comply with metro safety standards

Thermal Energy Storage Systems

To be competent, the user/individual on the job must be able to:

- PC11.** review the design and functioning of thermal energy storage systems at elevated and underground stations
- PC12.** verify the integration and efficiency of thermal energy storage systems in reducing energy consumption during peak hours

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the safety standards and protocols for metro railway E&M systems
- KU2.** the organizational policies on energy management and conservation
- KU3.** the maintenance schedules and compliance requirements for metro infrastructure
- KU4.** the principles of power distribution systems in metro stations
- KU5.** the functioning and maintenance of air-cooled chillers and fire suppression systems
- KU6.** the components and operation of ECS and TVS
- KU7.** the standards for earthing systems and electrical safety
- KU8.** the maintenance of water treatment, seepage, and sewage systems
- KU9.** the design and operation of thermal energy storage systems
- KU10.** the use of diagnostic tools and equipment for system inspections

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** effectively report and communicate findings to stakeholders
- GS2.** analyze system performance and identify areas of improvement
- GS3.** address issues in E&M systems promptly and effectively
- GS4.** document maintenance activities, inspections, and energy management plans
- GS5.** collaborate with maintenance teams and ensure task delegation
- GS6.** prioritize and schedule night visits and inspections efficiently
- GS7.** promote and ensure adherence to safety protocols
- GS8.** make informed decisions regarding maintenance and energy conservation
- GS9.** identify minor faults in systems during inspections
- GS10.** respond effectively to dynamic challenges during maintenance activities

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Night Visits for Maintenance Activities in Underground Stations</i>	3	3	-	1
PC1. check ongoing E&M maintenance activities during night visits to ensure adherence to standard operating procedures and safety regulations	-	-	-	-
PC2. verify the functionality and maintenance status of critical systems such as power distribution, lighting, ventilation, and escalators	-	-	-	-
<i>Energy Conservation and Management Practices</i>	3	3	-	1
PC3. monitor and analyze energy consumption trends to identify potential areas for conservation	-	-	-	-
PC4. follow appropriate energy-saving measures without compromising operational efficiency	-	-	-	-
<i>Power Supply Distribution and Cooling Systems</i>	4	3	-	2
PC5. evaluate the power supply distribution scheme at stations, ensuring compliance with DMRC standards and safety protocols	-	-	-	-
PC6. check the performance and maintenance of air-cooled chillers and aerosol fire suppression systems	-	-	-	-
<i>Earthing, Water Treatment, and Pumping Systems</i>	3	4	-	2
PC7. check earthing systems and ensure proper grounding to mitigate electrical hazards	-	-	-	-
PC8. coordinate the maintenance of water treatment plants (WTP), water supply pumps, seepage, and sewage systems	-	-	-	-
<i>Environmental Control Systems (ECS) and Tunnel Ventilation Systems (TVS)</i>	4	3	-	2
PC9. check ECS and TVS in underground stations for effective temperature and air quality management	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. ensure the ECS and TVS systems meet operational requirements and comply with metro safety standards	-	-	-	-
<i>Thermal Energy Storage Systems</i>	3	4	-	2
PC11. review the design and functioning of thermal energy storage systems at elevated and underground stations	-	-	-	-
PC12. verify the integration and efficiency of thermal energy storage systems in reducing energy consumption during peak hours	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0503
NOS Name	Monitor energy utilization and infrastructure maintenance in metro stations
Sector	Logistics
Sub-Sector	
Occupation	Electrical & Mechanical
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0504: Monitor and ensure safe escalator and elevator operations in metro systems

Description

This unit outlines the competencies required to monitor the escalator and elevator operations, ensuring safety, efficiency, and maintenance standards in metro railway systems.

Scope

The scope covers the following :

- Escalator Basics
- Visit to Escalator Simulator
- Elevator Basics
- Elevator Rescue Practical in Elevators
- Site Visit to See Maintenance Practices in Escalators
- Site Visit to See Maintenance Practices in Elevators
- Failure Analysis of Escalators and Elevators and RAM Parameters
- CBT of Lift and Escalator in Escalator Simulator Room

Elements and Performance Criteria

Escalator Basics

To be competent, the user/individual on the job must be able to:

- PC1.** identify the components, terminology, and specifications of an escalator, including the bottom pit, top pit, comb plate, skirt, chains, and track system
- PC2.** monitor commissioning activities of escalators to identify and address common issues such as alignment and safety calibration

Visit to Escalator Simulator

To be competent, the user/individual on the job must be able to:

- PC3.** familiarize with the functions of escalators, sensors, and systems and applicable safety measures using the simulator
- PC4.** perform analysis of escalator failures using the simulator and propose corrective actions

Elevator Basics

To be competent, the user/individual on the job must be able to:

- PC5.** identify the components, terminology, and specifications of an elevator, including traction machines, hoisting ropes, elevator cars, and rescue mechanisms
- PC6.** monitor elevator commissioning activities to ensure compliance with design specifications and resolve common operational issues

Elevator Rescue Practical in Elevators

To be competent, the user/individual on the job must be able to:

- PC7.** practice elevator rescue in the training facilities
- PC8.** follow the applicable standard procedures during mock elevator rescue drills and assess the performance of personnel to ensure readiness in emergencies

Qualification Pack

Site Visit to See Maintenance Practices in Escalators

To be competent, the user/individual on the job must be able to:

- PC9.** evaluate maintenance for escalators and ensure adherence to standard operating procedures
- PC10.** report discrepancies in maintenance and recommend improvements for escalator systems

Site Visit to See Maintenance Practices in Elevators

To be competent, the user/individual on the job must be able to:

- PC11.** check maintenance practices for elevators, including lubrication, alignment, and safety checks
- PC12.** verify compliance with maintenance schedules and document findings for elevator systems

Failure Analysis of Escalators and Elevators and RAM Parameters

To be competent, the user/individual on the job must be able to:

- PC13.** conduct a failure analysis to identify root causes and recurring issues in escalators and elevators
- PC14.** evaluate Reliability, Availability, Maintainability (RAM) parameters and suggest appropriate measures to improve system efficiency

CBT of Lift and Escalator in Escalator Simulator Room

To be competent, the user/individual on the job must be able to:

- PC15.** participate in computer-based training sessions to reinforce theoretical and practical knowledge of lift and escalator systems
- PC16.** use Computer-Based Training (CBT) modules to simulate and solve operational scenarios for escalators and elevators

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the principles of operation, terminology, components (bottom pit, top pit, comb plate, skirt, chains, track system), safety features, sensors, and commissioning challenges of escalator systems
- KU2.** the principles of operation, selection criteria, terminology, components (traction machine, hoisting rope, elevator car, rescue system), and commissioning challenges of elevator systems
- KU3.** the standard procedures for the maintenance of escalators and elevators, including lubrication, alignment, and safety checks
- KU4.** the techniques for identifying and addressing root causes of failures in escalators and elevators
- KU5.** the Reliability, Availability, and Maintainability (RAM) metrics for escalators and elevators
- KU6.** the safety protocols and procedures for elevator rescue and emergency response
- KU7.** the use of Computer-based Training (CBT) modules for lifts and escalators
- KU8.** the strategies for effective group discussions and team coordination
- KU9.** the codes, regulations, and guidelines relevant to escalator and elevator operations in metro systems
- KU10.** the preparation of maintenance reports and analysis documentation

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** assess escalator and elevator systems to identify and resolve operational issues
- GS2.** convey technical information effectively to team members and stakeholders
- GS3.** guide and oversee teams during commissioning and maintenance activities
- GS4.** devise solutions for failures and inefficiencies in escalator and elevator operations
- GS5.** monitor maintenance practices and ensure compliance with safety standards
- GS6.** utilize CBT tools and software for training and operational analysis
- GS7.** plan and execute site visits and practical sessions efficiently
- GS8.** work effectively in team settings
- GS9.** adjust to new technologies and methodologies in escalator and elevator systems
- GS10.** evaluate RAM parameters and propose improvements to enhance system performance

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Escalator Basics</i>	3	3	-	2
PC1. identify the components, terminology, and specifications of an escalator, including the bottom pit, top pit, comb plate, skirt, chains, and track system	-	-	-	-
PC2. monitor commissioning activities of escalators to identify and address common issues such as alignment and safety calibration	-	-	-	-
<i>Visit to Escalator Simulator</i>	2	2	-	1
PC3. familiarize with the functions of escalators, sensors, and systems and applicable safety measures using the simulator	-	-	-	-
PC4. perform analysis of escalator failures using the simulator and propose corrective actions	-	-	-	-
<i>Elevator Basics</i>	3	3	-	2
PC5. identify the components, terminology, and specifications of an elevator, including traction machines, hoisting ropes, elevator cars, and rescue mechanisms	-	-	-	-
PC6. monitor elevator commissioning activities to ensure compliance with design specifications and resolve common operational issues	-	-	-	-
<i>Elevator Rescue Practical in Elevators</i>	3	3	-	1
PC7. practice elevator rescue in the training facilities	-	-	-	-
PC8. follow the applicable standard procedures during mock elevator rescue drills and assess the performance of personnel to ensure readiness in emergencies	-	-	-	-
<i>Site Visit to See Maintenance Practices in Escalators</i>	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. evaluate maintenance for escalators and ensure adherence to standard operating procedures	-	-	-	-
PC10. report discrepancies in maintenance and recommend improvements for escalator systems	-	-	-	-
<i>Site Visit to See Maintenance Practices in Elevators</i>	2	2	-	1
PC11. check maintenance practices for elevators, including lubrication, alignment, and safety checks	-	-	-	-
PC12. verify compliance with maintenance schedules and document findings for elevator systems	-	-	-	-
<i>Failure Analysis of Escalators and Elevators and RAM Parameters</i>	3	3	-	1
PC13. conduct a failure analysis to identify root causes and recurring issues in escalators and elevators	-	-	-	-
PC14. evaluate Reliability, Availability, Maintainability (RAM) parameters and suggest appropriate measures to improve system efficiency	-	-	-	-
<i>CBT of Lift and Escalator in Escalator Simulator Room</i>	2	2	-	1
PC15. participate in computer-based training sessions to reinforce theoretical and practical knowledge of lift and escalator systems	-	-	-	-
PC16. use Computer-Based Training (CBT) modules to simulate and solve operational scenarios for escalators and elevators	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0504
NOS Name	Monitor and ensure safe escalator and elevator operations in metro systems
Sector	Logistics
Sub-Sector	
Occupation	Electrical & Mechanical
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per the assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/training centre based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 60% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N9901.Implement metro operations training and safety procedures	20	20	0	10	50	10
DMR/LSC/N0501.Monitor the operations and maintenance of E&M systems in metro railways	20	20	0	10	50	20
DMR/LSC/N0502.Monitor and maintain the underground and elevated station systems	20	20	0	10	50	20
DMR/LSC/N0503.Monitor energy utilization and infrastructure maintenance in metro stations	20	20	0	10	50	20
DMR/LSC/N0504.Monitor and ensure safe escalator and elevator operations in metro systems	20	20	0	10	50	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	120	130	0	50	300	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
OJT	On the Job Training
VRV	Variable Refrigerant Volume
ECS	Environmental Control System
TVS	Tunnel Ventilation System
BMS	Building Management System
NT-SCADA	Networked Supervisory Control and Data Acquisition
UPS	Uninterruptible Power Supply
DG	Diesel Generator
LT	Low Tension
ACB	Air Circuit Breaker
SCADA	Supervisory Control and Data Acquisition
SMI	Special Maintenance Instructions
SOI	Standard Operating Instructions

Logo
Not
Available



Qualification Pack

E&M	Electrical and Mechanical
RAM	Reliability, Availability, and Maintainability
CBT	Computer-Based Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.